5.5.1 Setting an exploration schedule for Upper Confidence Bound

This exercise, implemented in `CH05/02 - Exercise 1.ipynb`, discusses a strategy of adaptively setting the value of the tradeoff parameter β of the Upper Confidence Bound (UCB) policy. As mentioned in the section on UCB, the performance of the policy heavily depends on this parameter, but it's not obvious how we should set its value. A value might work well on some objective functions but poorly on others.

Bayesian optimization practitioners have noticed that as we collect more and more data, UCB can become too exploitative. This is because as the size of our training data set increases, we gain more knowledge about the objective function, and our uncertainty about the predictions made by the Gaussian process decreases. This means that the credible interval produced by the Gaussian process will become tighter, making the upper bound that makes up the acquisition score used by UCB closer to the mean prediction.

However, if the UCB acquisition score is similar to the mean prediction, then the policy is exploiting, as it will only query data points with high predictive means. This phenomenon shows that the more data we have observed, the more explorative we should be with UCB. Here, a natural way of gradually encouraging more exploration from UCB is to slowly increase the value of the tradeoff parameter β, which is what we will learn to do in this exercise. There are multiple steps:

1. Recreate the Bayesian optimization (BayesOpt) loop in `CH06/02 - Exercise 1.ipynb`, which uses the one-dimensional Forrester function as the optimization objective.
2. We aim to slowly increase the value of the tradeoff parameter by multiplying it with a constant at each iteration of the loop. That is, at the end of each iteration, we need to update the parameter with `beta *= multiplier`.

   Say we want to start out having a value of 1, and to have a value of 10 at the end of the search (the tenth iteration). What is the value the multiplier for needs to have?
3. Implement this scheduling logic and observe the resulting optimization performance.
   A. Specifically, even though this version of UCB starts out with = *1*, does it get stuck at a local optimum like the version where the parameter is fixed at 1?

5.5.2 Bayesian optimization for hyperparameter tuning

This exercise, implemented in `CH05/03 - Exercise 2.ipynb` applies BayesOpt to an objective function that simulates the accuracy surface of a support-vector machine model in a hyperparameter tuning task. The x-axis denotes the value of the penalty parameter C, while the y -axis denotes the value for the RBF kernel parameter γ. See the exercises in Chapter 3 and Chapter 4 for more detail. There are multiple steps:

1. Recreate the Bayesian optimization loop in `CH06/03 - Exercise 2.ipynb`, including the outer loop that implements repeated experiments.
2. Run the Upper Confidence Bound (UCB) policy, setting the value of the tradeoff

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